5.1 MI

$\boxed{P(n)} \longrightarrow$

- Verify for $n = 1$
- Assume $P(n) \longrightarrow$ true for $K \in \mathbb{Z}^+$
- $P(K+1) \longrightarrow$ must be true if $P(K)$ is true

ex $\rightarrow P(n) = \boxed{1} + 2 + 3 + \cdots + n = \left(\frac{n(n+1)}{2}\right)$

1. $P(1) \rightarrow L.H.S = 1 \qquad R.HS = \frac{1(1+1)}{2} = 1$

2. $P(K) \rightarrow K \in \mathbb{Z}^+ \rightarrow \boxed{1+2+3+\cdots+K} = \frac{K(K+1)}{2}$ (Assume true)

3. $P(K+1) \rightarrow 1+2+3+\cdots+(K+1) = \boxed{\frac{(K+1)(K+2)}{2}}$

$$1+2+3+\cdots+K+(K+1)$$
$$= \frac{K(K+1)}{2} + (K+1)$$
$$= \frac{K(K+1)+2(K+1)}{2}$$
$$= \frac{(K+1)(K+2)}{2}$$

ex, $n \in \mathbb{Z}^+$ $n < 2^n$

$n = 1 \rightarrow 1 < 2^1 \Rightarrow 1 < 2$ ✓

$n = k \in \mathbb{Z}^+ \rightarrow k < 2^k$ (Assume true)

$n = (k+1) \in \mathbb{Z}^+ \rightarrow (k+1) < 2^{k+1}$

$2 < 3$
$2 < 5$ } $3 < 5$

$\boxed{k < 2^k}$

$k+1 < 2^k + 1$

$k+1 < 2^k + 2^k$

$k+1 < 2 \cdot 2^k$

$\boxed{k+1 < 2^{k+1}}$

MI $n < 2^n$ ✓

5.3 Recursive Defn

$$\sum_{k=0}^{n} a_k = \underbrace{\boxed{a_0 + a_1 + a_2 + \cdots + a_{n-1}}} + a_n$$

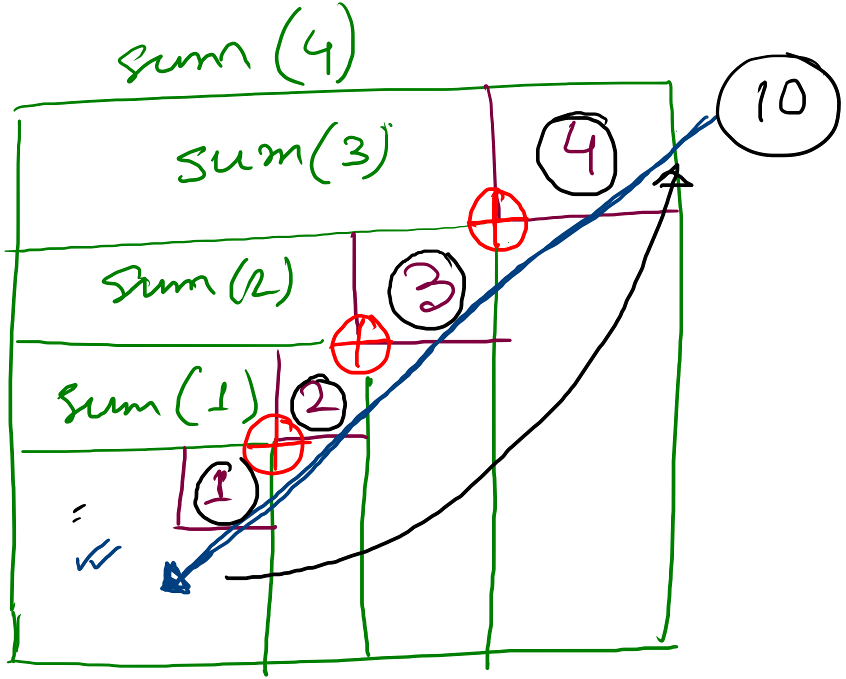
$$\sum_{k=0}^{n-1} a_k + a_n$$

$$\sum_{k=0}^{0} a_k = a_0$$

| 1 | 2 | 3 | 4 | → 10

sum(n) $\sum_{k=1}^{4} a_k = 4 + \sum_{k=1}^{3} a_k =$

$$3 + \boxed{\sum_{k=1}^{2} a_k}$$



5.4 → RA

⊛ factorial

$5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 4! \cdot 5$

$4! = 1 \cdot 2 \cdot 3 \cdot 4$

$n! = (n-1)! \cdot n$

$1! = 1$

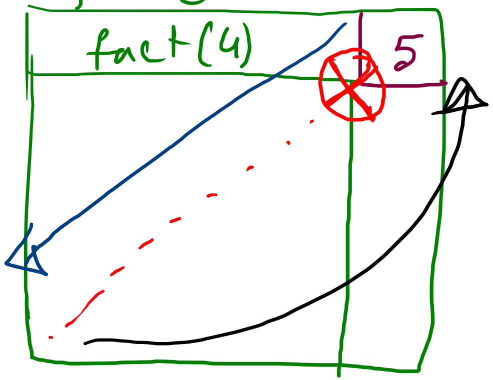
$0! = 1$

procedure fact(n)

if $n = 0$ return 1

else return $n \cdot$ fact(n-1)

fact(5)



⊛ power

$pow(a, n) \equiv a^n$

$2^3 = 2 \cdot 2^2$

$2 \cdot 2^1$

$\boxed{2^1 = 2 \cdot 2^0}$

procedure pow(a,n)

if n==0 return 1;

else return a. pow(a,n-1)

① fibonacci

i → 1 2 3 4 5

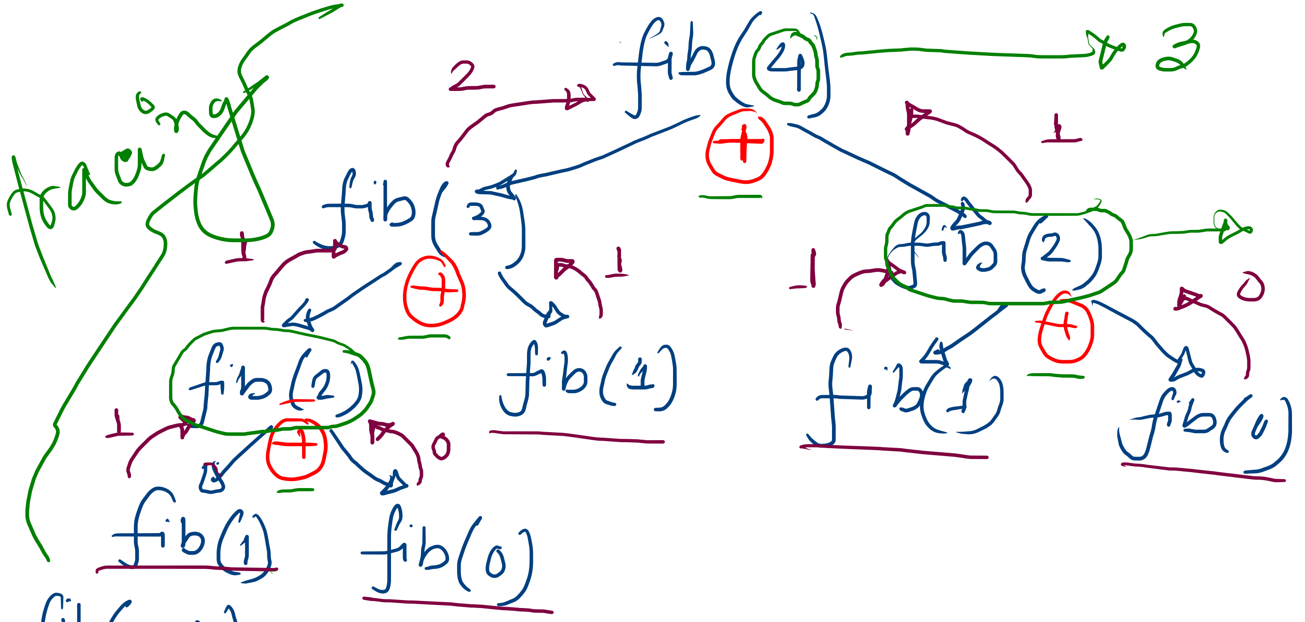
(n-2) (n-1)

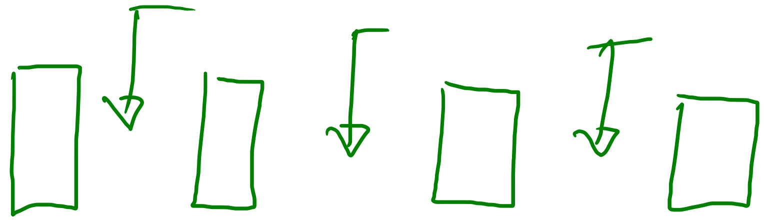
| 0 | 1 | 1 | 2 | 3 | . . . . .

fib(n) → $(n+1)^{th}$

if n==0 return 0
elif n==1 return 1
else return. fib(n-1)+fib(n-2)

tracing





$n = (n+1)^{th}$ fib no.

DP